

Manufacturer Channel Encroachment and Evolution in E-Platform Supply Chain: An Agent-Based Model

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1. RESEARCH OBJECTIVE & SYSTEM ARCHITECTURE

Multi-agent model simulating manufacturer, e-platform, and consumer agents using genetic algorithm pricing adjustments and market evolution.

2. BENCHMARK DATASETS & EMPIRICAL EVALUATION

Agent-based simulation across 100 periods with synthetic agent populations.

Key Numerical Findings & Empirical Proofs:
Demonstrates that manufacturer channel encroachment reduces e-platform profitability, while consumer quality preference discourages encroachment.

3. CRITICAL ARCHITECTURAL FLAWS & METHODOLOGICAL LIMITATIONS

Identified System Failure Modes:
Synthetic toy agents driven by rigid utility formulas; zero cognitive reasoning or negotiation dialogue; uncalibrated against real-world market datasets.

4. RESEARCH GAP & RELEVANCE TO CAG-DELPHI

This work exemplifies the *Unconditional Consultation Dilemma* observed across the 15-paper corpus. By failing to incorporate epistemic uncertainty self-gating prior to multi-agent activation, the architecture incurs severe token inflation and exposes confident primary solutions to debate degeneration noise. In CAG-Delphi, this paper serves as an empirical anchor for diversity-governed consensus and comparative baseline evaluation.

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